

As usual, ω is minimal such that $T_t = T_\omega$ and $\bar{\omega}$ is minimal such that $S_t = S_{\bar{\omega}}$. T_t , for $t < \omega$, has a $\mathbb{1}$ -edge. (Indeed, otherwise the Levitt decomposition of T_t has components that are dense or simplicial with $\mathbb{Z}$ -edges and so $T_t \rightarrow T_\omega$ is an isometry (needs reference)) If $\bar{\omega} \leq t < \omega$ then $\Psi(T_t) \not\prec T_t \searrow S$ and so $T_t \in FS_V$ by definition. If $\bar{\omega} = \omega$ then eventually V is visible in S_t for $t < \omega$ (needs reference) and so $T_t \searrow S_t \searrow S$. Hence $\Psi(T_t) \not\prec T_t \searrow S$. $\square$

Proof. (Old.) Using Lemma 20.2, ω is the minimal t such that V is elliptic in T_t and that, for all $t < \omega$, T_t has a $\mathbb{1}$ -edge. Give T_ω the transverse cover, call it $\mathcal{T}_\omega$, in Lemma 12.5 (see [13, Lemma 5.8]). Using the terminology of Proposition 20.5, ω is also minimal so that V is elliptic in S_ω where S_t denotes the skeleton of the transverse cover $\mathcal{T}_t$ there. Indeed, $\mathcal{T}_\omega$ has no component vertices with stabilizer V (and so the same is true for $\mathcal{T}_t$ with $t < \omega$) and $\mathcal{T}_t$ has no point vertices with stabilizer V since T_t has no points with stabilizer V for $t < \omega$. There is a point vertex in $S_{\omega-}$ (see Remark 20.6) stabilized by V . It follows from Lemma 20.10 that there is $\omega' < \omega$ such that V is visible in S_t for $t \in [\omega', \omega)$. We have that T_t is a graph of actions with underlying graph of actions given by S_t . Consider the resulting graph of actions on T_t with underlying graph the $\bar{S}_t$ that results from collapsing edges of S_t not in the subgraph defining V . By construction $\bar{S}_t$ has a point v with stabilizer V . T_t has one edges and, since V is filling, they must all be in the vertex action at v (and its $\mathbb{F}$ -orbit). Let $\overline{S_t + T_t}$ denote the result of further collapsing the vertex actions not at v and replacing the vertex action T_v at v with its image under $T_v \subset T_t \rightarrow \bar{T}_L$ where $\bar{T}_L$ is the quotient of T_t obtained from its Levitt decomposition by collapsing all but $\mathbb{1}$ -edges. Note that, since T_v is closed in T_t , this image is a closed subtree of a simplicial $\mathbb{1}$ -tree [11, Lemma 4.10]. We have

$$\overline{S_t + T_t} \not\prec S_t + T_t \searrow T_t$$

By construction $\Psi(\overline{S_t + T_t}) \in FS_V$ and so $\Psi(T_t)$ is (coarsely) in FS_V . $\square$

Add lemma about blowing up visible graph using $\mathbb{1}$ -edges is in FS_V ?

20.3 Case (b)

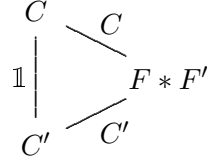
Here I am.

Some stuff to be inserted around Proposition 20.5.

Somewhere put lemma about folding paths are quasigeodesics

Here is an example of a transverse cover on $T := (F_4)_{*1} \in FS(F_5)$ whose skeleton S has infinitely generated edge groups.

Example 20.12. Consider $T \not\prec \hat{S} \searrow S$ where $\hat{S}$ has Bass-Serre diagram



F and F' each have rank two, $C < F$ and $C' < F'$ are infinitely generated, S is obtained from $\widehat{S}$ by collapsing all $\mathbb{1}$ -edges, and T is obtained from $\widehat{S}$ by collapsing all other edges. A component of the transverse cover is obtained by taking a component of the union of the $\mathbb{1}$ -edges in the tree with presentation $\widehat{S}$ and projecting into T .

Lemma 20.13. *Suppose T has a Bass-Serre diagram with finitely many edges. Then the Bass-Serre diagram S for the skeleton of any transverse cover $\mathcal{T}$ on T has finitely many edges.*

Proof. Every edge of T crosses finitely many components of $\mathcal{T}$. Let T' be the result of subdividing T at vertices in more than one component of $\mathcal{T}$. There are finitely many orbits of vertices in T' and hence finitely many orbits of directions at each point vertex of S . Since S is bipartite and we conclude that S has only finitely many orbits of edges. $\square$

Lemma 20.14. *Suppose $T_0 \xrightarrow{f} T \searrow \widehat{T}$ where f is a morphism that is injective on the union of $\mathbb{Z}$ -edges of T_0 . Then the pullback $\widehat{f} : \widehat{T}_0 \rightarrow \widehat{T}$ is also injective on the union of $\mathbb{Z}$ -edges.*

Proof. Suppose $x, y \in \widehat{T}_0$ are in $\mathbb{Z}$ -edge and have the same $\widehat{f}$ image. Complete. $\square$

^(1:b) **Lemma 20.15.** *Suppose T_t , $t \in [0, \infty)$ is a tame greedy optimal folding path such that T_0 is simplicial with only $\mathbb{1}$ -edges and T_ω is in $\overline{\mathcal{O}}$ where $\omega < \infty$ is minimal such that $T_\omega = T_t$, $t \geq \omega$. Suppose also that there is $V \in \mathbf{W}$ such that $V|T_\omega$ has dense orbits and collapses to a $(V, \mathcal{F}_V)$ -averse tree $\overline{V|T_\omega}$. Then, in $\overline{FS}$, $\Phi(T_t)$ converges to $[\overline{V|T_\omega}] \in \partial FC_V \subset \partial FS$; see Proposition 9.4.*

Use notation there.

Proof. (New.) T is a generalized black hole and so a graph of actions over a Bass-Serre tree in FC with V an edge stabilizer. Let $\overline{T}$ be the graph of actions over the same tree but with $\overline{V|T}$ replacing $V|T$. We have a collapse map $T \searrow \overline{T}$ and so a collapse diagram

$$\begin{array}{ccccc}
T_0 & \longrightarrow & T_{\bar{\omega}} & \longrightarrow & T_{\omega} = T \\
& \searrow & & \searrow & \searrow \\
& & \bar{T}_0 & \longrightarrow & \bar{T}_{\bar{\omega}} = \bar{T}
\end{array}$$

By construction, $V|\bar{T}$ is a $(V, \mathcal{F}_V)$ -averse tree and so $\Psi(V|\bar{T}) := \xi \in \partial FS_V \subset \partial FS$ is defined and if $\Psi(\bar{T}_t)$, $t < \bar{\omega}$, is eventually in FS_V then $\Psi(\bar{T}_t)$ (and so also $\Psi(T_t)$) converges to ξ in $\bar{FS}$. (needs reference) The remainder of the proof is to show that $\Psi(\bar{T}_t)$ is eventually in FS_V .

$\bar{T}$ is a graph of actions with V a vertex stabilizer of its underlying Bass-Serre tree $S \in FC$. (needs reference) Hence $\bar{T} \swarrow \hat{\bar{T}} \searrow S$ where $\hat{\bar{T}}$ is obtained from the graph of actions structure on $\bar{T}$ by declaring length zero edges of S to have positive length. There is then a pullback diagram

$$\begin{array}{ccccc}
\hat{\bar{T}}_0 & \longrightarrow & \hat{\bar{T}}_{\bar{\omega}'} & \longrightarrow & \hat{\bar{T}}_{\hat{\omega}} = \hat{\bar{T}} \\
& \searrow & & \searrow & \searrow \\
\bar{T}_0 & \longrightarrow & \bar{T}_{\bar{\omega}'} = \bar{T} & \longrightarrow & \bar{T}_{\hat{\omega}} = \bar{T}
\end{array}$$

Recall that the bottom line of the pullback diagram is $\bar{T}_0 \rightarrow \bar{T}_{\bar{\omega}}$ reparametrized (using a nondecreasing map of the reals). The subscript $\bar{\omega}'$ denotes the first time $\bar{T}_t = \bar{T}$.

- Note that, when restricted to V , the rightmost parallelogram in the pullback diagram consists of isometries. (Indeed, a map that is both a collapse and a morphism is an isometry.) Hence, $\hat{\bar{T}}_{\bar{\omega}}$ (the middle tree in the top line of the pullback diagram) is a surface type graph of actions rel V . Indeed, the proof of Proposition 19.12 generalizes to tame trees.

Go back and
restate prop for
tame trees

- Finally consider the collapse diagram

$$\begin{array}{ccc}
\hat{\bar{T}}_0 & \longrightarrow & \hat{\bar{T}}_{\hat{\omega}} \\
& \searrow & \searrow \\
& S_0 & \longrightarrow S
\end{array}$$

induced by the top row of the pullback diagram and $\hat{\bar{T}}_{\hat{\omega}} \searrow S$. Recall that $S \in FC$ and has V as a vertex stabilizer. Since $V|\hat{\bar{T}}_{\bar{\omega}'}$ has dense

orbits, V is a vertex stabilizer in its collapse $S_{\bar{\omega}'}$. There are two cases: there is $\epsilon > 0$ such that, for $\bar{\omega}' - \epsilon < t < \bar{\omega}'$, either V is elliptic in S_t or V is visible in S_t . (needs reference) In either case there is a collapse of $\bar{S}_t$ of S_t with V a vertex stabilizer. For t in this range we have $\Psi(\bar{T}_t) \searrow \bar{T}_t \searrow \hat{\bar{T}}_t \searrow \bar{S}_t$, ie $\bar{\Psi}(T_t) \in FS_V$ as promised.

□

Proof. (Old.) T is a generalized black hole and so a graph of actions over a Bass-Serre tree in FC with V an edge stabilizer. Let $\bar{T}$ be the graph of actions over the same tree but with $\bar{V}|T$ replacing $V|T$. By (needs reference) we may assume that $\bar{V}|\bar{T}$ is mixing. We have a collapse map $T \searrow \bar{T}$ and so a collapse diagram

$$\begin{array}{ccccc} T_0 & \longrightarrow & T_{\bar{\omega}} & \longrightarrow & T_{\bar{\omega}} = T \\ & \searrow & & \searrow & \searrow \\ & & \bar{T}_0 & \longrightarrow & \bar{T}_{\bar{\omega}} = \bar{T} \end{array}$$

$\bar{T}$ is a graph of actions with V a vertex stabilizer of the underlying Bass-Serre tree $S \in FC$. Hence $\bar{T} \searrow \hat{\bar{T}} \searrow S$ where $\hat{\bar{T}}$ is obtained from the graph of actions structure on S by declaring length zero edges to have positive length. There is then a pullback diagram

$$\begin{array}{ccccc} \hat{\bar{T}}_0 & \longrightarrow & \hat{\bar{T}}_{\bar{\omega}'} & \longrightarrow & \hat{\bar{T}}_{\hat{\omega}} = \hat{\bar{T}} \\ & \searrow & & \searrow & \searrow \\ \bar{T}_0 & \longrightarrow & \bar{T}_{\bar{\omega}'} = \bar{T} & \rightarrow & \bar{T}_{\hat{\omega}} = \bar{T} \end{array}$$

Recall that the bottom line of the pullback diagram is $\bar{T}_0 \rightarrow \bar{T}_{\bar{\omega}}$ reparametrized (using a nondecreasing map of the reals). The subscript $\bar{\omega}'$ denotes the first time $\bar{T}_t = \bar{T}$.

- $\bar{T}$ is not a black hole: Suppose $\bar{T}$ is a W -black hole and so is a graph of actions of surface type. $V|T$ inherits a surface type graph of actions structure. Since $V|T$ is mixing, it is a single surface. Consider the splitting in FC given by boundaries of all surfaces. It has V and W as vertex stabilizers, contradicting (needs reference).

- In the pullback diagram, $\widehat{T}_{\overline{\omega}}$ (the middle tree on the top line) is a surface type graph of actions rel V . Indeed, generalize proof of Proposition 19.12. Note that, when restricted to V , the triangle in the diagram consists of isometries. In particular, $\widehat{T}_{\overline{\omega}}$ has no $\mathbb{1}$ -edges, $\widehat{T}_{\overline{\omega}} \rightarrow \widehat{T}_{\widehat{\omega}}$ only folds $\mathbb{Z}$ -edges, and the Bass-Serre tree underlying $\widehat{T}_{\overline{\omega}}$ has V as a vertex stabilizer and no edges of length zero. It follows that $\Psi(\widehat{T}_t)$, $t < \overline{\omega}$ is eventually in FS_V . (In this range, $\widehat{T}_t$ has a $\mathbb{1}$ -edge since $\overline{T}_t$ does; see the $\mathbb{Z}$ -injective discussion below.)
- Viewing FS_V as a convex subset of FS , the limit of $\Psi(\widehat{T}_t)$ is the same as its projection to FS_V , which is the same as restriction for a tail of this folding path. Since $V|\widehat{T}_t$ converges to $V|T$, a tail of $\Psi(\widehat{T}_t)$ converges to the point ξ of ∂FS corresponding to $[V|\overline{T}] = [V|T] \in \partial FC_V$ (needs reference). Since $\Psi(\widehat{T}_t) = \Psi(\overline{T}_t)$ for t in our range, $\Psi(\overline{T})$ has the same limit.
- Say a morphism $f : R \rightarrow R'$ is $\mathbb{Z}$ -injective if $f|(Fix_E(R))$ is injective for all infinite cyclic $E < \mathbb{F}$.
 - $T_t \rightarrow T$ is $\mathbb{Z}$ -injective: Indeed, $T_0 \rightarrow T$ is $\mathbb{Z}$ -injective and T_t is greedy.
 - $\overline{T}_t \rightarrow \overline{T}$ is $\mathbb{Z}$ -injective: Indeed, $T \rightarrow \overline{T}$ is $\mathbb{Z}$ -injective and so $\overline{T}_t \rightarrow \overline{T}$ is also.
 - $\widehat{T}_t \rightarrow \widehat{T}$ need not be $\mathbb{Z}$ -injective. This cost us the extra work above.

The key point is that $\mathbb{Z}$ -injective morphisms can't fold $\mathbb{Z}$ -edges, exhibiting the presence of a $\mathbb{1}$ -edge unless the morphism is an isometry.

- Using the collapse diagram, that $\overline{T}$ is not a black hole, and Proposition 20.8, we have that $\Psi(T_t)$, $t < \omega$ also converges to ξ in $\overline{FS}$.

□

Proof. (Older.) (1) T is a generalized black hole and so is a graph of actions as in Proposition 19.12. Let $\widehat{T}$ be the graph of actions obtained from T by declaring each edge of zero length in the skeleton to have length one. We may assume that all $\mathbb{Z}$ -edges of T are included in its underlying Bass-Serre presentation.

$$T \swarrow \widehat{\widehat{T}} \searrow S$$

Restrict to V

$$\begin{array}{ccccccc} \widehat{\widehat{T}}_0 & \longrightarrow & \widehat{\widehat{T}}_\omega & \longrightarrow & \widehat{\widehat{T}}_{\widehat{\omega}} & = & \widehat{\widehat{T}} \\ & \searrow & & \searrow & \searrow p & & \searrow \\ & & T_0 & \longrightarrow & T_\omega = T & & S \end{array}$$

- $\widehat{\widehat{T}}_0 \rightarrow \widehat{\widehat{T}}$ is injective on $Fix_E(T)$ where E is a nontrivial edge stabilizer in $\widehat{\widehat{T}}_0$. Indeed, suppose $(t, \hat{t}) \neq (t', \hat{t}')$ are in $Fix_E(\widehat{\widehat{T}}_0)$ and have the same image in $\widehat{\widehat{T}}$. There are cases.
 - $(t \neq t')$: Then $[t, t'] \subset T_0$ is fixed by E , contradiction.
 - $(t = t')$: By construction, the pullback is injective on fibers.
- $\widehat{\widehat{T}}_t$ for $t < \widehat{\omega}$ has a $\mathbb{1}$ -edge. Indeed, otherwise, by the previous item, $\widehat{\widehat{T}}_t \rightarrow \widehat{\widehat{T}}$ is an isomorphism on $\mathbb{Z}$ -edges. Give $\widehat{\widehat{T}}_t$ the resulting graph of actions structure. If $\widehat{\widehat{T}}_t \neq \widehat{\widehat{T}}$ then the restriction to some vertex action is not an isomorphism. Since all vertex actions have dense orbits, this is a contradiction.
- By Proposition 20.8, $\{\Psi(\widehat{\widehat{T}}_t) | \omega \leq t < \widehat{\omega}\}$ has uniformly bounded diameter in FS .
- $\Psi(\widehat{\widehat{T}}_t)$ is eventually in FS_V . Indeed, V is the stabilizer of a vertex of the skeleton corresponding to a cover element. Following Proposition 20.5, consider the induced transverse cover on $\widehat{\widehat{T}}_t$ for $t < \widehat{\omega}$. Let $\widehat{\widehat{S}}_t$ denote the skeleton. There are cases.
 - $(\widehat{\widehat{S}}_t = \widehat{\widehat{S}}_{\widehat{\omega}} \text{ for some } t < \widehat{\omega})$: Then $\widehat{\widehat{T}}_t \rightarrow \widehat{\widehat{T}}$, for t near $\widehat{\omega}$ folds the component with stabilizer V and is constant on other components. In particular, $\widehat{\widehat{T}}_t$ is in FS_V .
 - $(\widehat{\omega} \text{ is minimal such that } \widehat{\widehat{S}}_t = \widehat{\widehat{S}})$: Using the notation of Proposition 20.5 and Remark 20.6,

By Lemma 20.11, and, by Proposition 20.8, the Ψ -image of the path from $T_{\widehat{\omega}}$ and T_ω has uniformly bounded diameter.

(2) By Camille and Dowdall-Taylor (needs reference) and using (1), T_t converges to $[V|T] \in \partial FC_V \subset \partial FS$. $\square$

20.4 Case (a)

Lemma 20.16. *Suppose T_t , $t \in [0, \infty)$ is a greedy folding path in $\overline{\mathcal{O}}$ such that T_0 is simplicial with only 1-edges and $\omega < \infty$ is minimal such that $T_\omega = T_t$, $t \geq \omega$. Suppose also that $T \searrow \overline{T}$ where $\overline{T}$ has a transverse cover with skeleton S in FZ' . Then $\Phi(T_t)$ converges to $\Phi(S)$.*

Proof. (New.) We have a collapse diagram

$$\begin{array}{ccc} T_0 & \longrightarrow & T \\ & \searrow & \searrow \\ & \overline{T}_0 & \longrightarrow \overline{T} \end{array}$$

a pullback diagram

$$\begin{array}{ccc} \widehat{T}_0 & \longrightarrow & \overline{T} + S \\ & \searrow & \searrow \\ \overline{T}_0 & \longrightarrow & \overline{T} \end{array}$$

and another collapse diagram

$$\begin{array}{ccc} \widehat{T}_0 & \longrightarrow & \overline{T} + S \\ & \searrow & \searrow \\ S_0 & \longrightarrow & S \end{array}$$

Let $\widehat{\omega}$ be minimal such that $\widehat{T}_t$ collapses to S .

The resulting folding path to S has Ψ -image converging to $\Psi(S)$. Since neither S nor $\overline{T}$ is a black hole, we conclude the Ψ -image of T_t converges to $\Psi(S)$. $\square$

Proof. (Old.) The proof is similar to that of Lemma 20.15. Consider the universal constructions resulting from

$$\begin{array}{ccccc} T_0 & \longrightarrow & T & & \overline{T} + S \\ & & \searrow & \nearrow & \searrow \\ & & \overline{T} & & S \end{array}$$

The resulting folding path to S has Ψ -image converging to $\Psi(S)$. Since neither S nor $\bar{T}$ is a black hole, we conclude the Ψ -image of T_t converges to $\Psi(S)$. $\square$

This finishes the proof of Proposition 20.1.

Reconcile 1,2,3
with a,b,c

20.5 Limits of sequences

Rewrite to conform with updated trichotomy (or is it dichotomy. Lemma ???).

Recall (Lemma 21.1) that if T_t is a folding path in $\bar{\mathcal{O}}$ where each T_t has a $\mathbb{1}$ -edge, then $\Psi(T_t)$ is a uniform unparametrized quasigeodesic in $FS(\mathbb{F}, \mathcal{F})$. In particular, $S := \lim \Psi(T_t) \in \bar{FS}$ is coarsely well-defined.

(1:paths to sequences) **Lemma 20.17.** *Suppose*

- $T_t \rightarrow T$, $t \in [0, \omega]$, is the canonical optimal folding path guided by $T_0 \rightarrow T_\omega$;
- ω is minimal such that $T_t = T_\omega$;
- T_0 is simplicial with only $\mathbb{1}$ -edges; and
- T_i is a sequence in $\mathcal{O}$ converging to T .

Then $\Psi(T_i) \xrightarrow{r} S := \Psi(T)$.

Proof. We show that every subsequence has a deeper subsequence converging to S . Let τ_{ij} , $i < j$ and τ_i be greedy folding paths such that:

bad notation: T_t
a folding path to
 T , T_i a sequence
to T . Change T_i
to T'_i

- τ_{ij} is guided by optimal $T_{ij} \rightarrow T_j$ where T_{ij} is in the closed simplex Σ_i (independent of j) of an element of $\mathcal{O}$ that shares a collapse with T_i . In particular, $d_{FS}(\Psi(T_{ij}), \Psi(T_i)) \leq 1$.
- τ_i is guided by $\hat{T}_i \rightarrow T$, where $\hat{T}_i$ is in Σ_i ; and
- τ_{ij} converges to τ_i , ie $\tau_{ij}(t) \rightarrow \tau_i(t)$ as $j \rightarrow \infty$.

See for example [11, Proof of Proposition 8.12] for the existence of such paths. By Lemma 20.2 all these paths are in $\bar{\mathcal{O}}$.

Suppose first that $S \in FS$. Choose $\epsilon > 0$ such that $\Phi(\tau_i(t_i - \epsilon))$ is uniformly close to S where t_i is the first time t such that $\tau_i(t)$ reaches T . We have that $(\tau_{ij}(t_i - \epsilon))_j$ converges to $\tau_i(t_i - \epsilon)$. Hence, for all large j , the geodesic from $\Psi(T_i)$ to $\Psi(T_j)$ passes uniformly close to S .

The case where $S \in \partial FS$ is similar. $\square$

The following proposition follows easily from Lemma 20.18 and Section 20.1.

Proposition 20.18. *Suppose the sequence $T_i \in \mathcal{O}$ converges to $T \in \overline{\mathcal{O}}$.*

- (1) *Suppose T is as in Proposition 13.3(2), ie there is $W \in \mathbf{W}$ such that $W|T$ has dense orbits and collapses to a $(W, \mathcal{F}_W)$ -averse tree T_W . Then, in $\overline{FC}_W$, $\Psi_W(W|T_i)$ converges to the boundary element $\Psi_W(T_W)$ and, in $\overline{FS}$, $\Psi(T_i)$ converges to the corresponding boundary element.*
- (2) *Suppose T is as in Proposition 13.3(1), ie T is a V -black hole for $V \in \mathbf{W}$. Then $\Psi(T_i)$ is eventually in $FS_V \subset FS$.*
- (3) *Suppose T is as in Proposition 13.3(3). Then $\Psi(T_i) \xrightarrow{r} \Psi(S)$.*

21 Random Lemmas

$\langle 1:\text{quasigeo} \rangle$ **Lemma 21.1.** *if T_t is a folding path in $\overline{\mathcal{O}}(\mathbb{F}, \mathcal{F})$ where each T_t has a $\mathbb{1}$ -edge, then $\Psi(T_t)$ is a uniform unparametrized quasigeodesic in FS . [grep from earlier version](#)*

Lemma 21.2. *Suppose*

- $T_t, 0 \leq t \leq \omega$, *is the canonical optimal folding path guided by $T_0 \rightarrow T_\omega$;*
- $T_0, T_\omega \in \overline{\mathcal{O}}$; *and*
- $Fix_E(T_0) \rightarrow Fix_E(T_\omega)$ *is injective for all maximal cyclic E .*

Then $T_t \in \overline{\mathcal{O}}, 0 \leq t \leq \omega$.

Proof idea. Set $t_0 := \inf\{t | Fix_E(T_t) \rightarrow Fix_E(T_\omega) \text{ is not injective}\}$ is zero. Let $\overline{T}_t$ be obtained from T_t by collapsing the dense Levitt components. There is an induced *generalized folding path* $\overline{T}_t$. Each $\overline{T}_s \rightarrow \overline{T}_t, s \leq t$ is a collapse followed by a morphism. If s and t are close then the subsets of $\overline{T}_s$ that are collapses are initial segments of $\mathbb{1}$ -edges. Note that $t_0 = \inf\{t | Fix_E(\overline{T}_t) \rightarrow Fix_E(\overline{T}_\omega) \text{ is not injective}\}$. [More words.](#)

□

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